

Victor Barres, PhD

Research Scientist · Conversational Agents & Evaluation

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Experience

Mercor

Research Scientist

Jun 2026 – Present

San Francisco, CA

- Founding member of the **APEX research team** ([AI Productivity Index](#)), measuring how capable AI is at economically valuable work.
- Research on conversational AI agents inside organizations — how they work with people, how they are transforming work itself, and how to measure that well enough to understand it.

Sierra

AI Researcher

Sep 2024 – May 2026

San Francisco, CA

- Led the τ -**Bench family** of agent benchmarks — research direction, codebase, public leaderboard ([taubench.com](#)), and ecosystem of extensions: τ^2 -Bench (dual control), τ -Knowledge (knowledge retrieval), τ -Voice (full-duplex voice), and τ^3 -Bench (the combined release with community-contributed task and code improvements). Three papers from the family accepted to **ICML 2026**, with τ^2 -Bench as an **oral**. Continued contributing to the line after leaving: Hyper- τ -Bench (agents that build agents), τ -Multilingual, and τ -Elicitation (2026).
- Kickstarted Sierra's internal **agent evaluation system**: set up the proof of concept and did the original work on user-simulation-driven stress testing for pre-release agent hardening across deployed customer-service domains.

Elemental Cognition

Senior Researcher (NLP)

Jun 2023 – Jul 2024

Remote

- Developed core algorithms for EC's hybrid AI architecture, integrating LLMs with formal reasoning engines.
- Led the integration of multi-round RAG with formal models of causation for knowledge-intensive multi-hop QA, published as *From Generating Answers to Building Explanations* at NAACL 2025 (Industry).
- Built conversational agents for two reasoning-heavy settings: a logic-programming assistant that elicits and translates user requirements into a logic program, and an ASP-reasoner-backed constraint-solving agent.

Uniphore.ai

Principal NLP Scientist & NLP Research Lead, previously Senior NLP Scientist (2019–2021)

Dec 2019 – Jun 2023

Palo Alto, CA

- Built and led Uniphore's NLP Lab (7+ scientists). Set short-, mid-, and long-term research strategy for understanding human-human spoken conversation in the contact-center domain.
- Shipped foundational and applied work across LLM benchmarking, prompt engineering, structured knowledge grounding, summarization, semantic parsing, dialogue-act and dialogue-flow modeling, and real-time caller-intent recognition.
- Kickstarted the team, deployed first NER / redaction / call-driver models, and stood up cloud infrastructure and CI/CD for production NLP.

Astound.ai

AI Scientist (NLP)

Sep 2017 – Dec 2019

Menlo Park, CA

- Core member of the team that built, from the ground up, Astound's domain-expert conversational agent for IT / HR / Finance enterprise self-service.
- Led the neuro-symbolic side of the agent's NLP stack — word / sentence embeddings, semantic parsing, knowledge-graph embeddings, and semantic inference — integrated with a modular dialogue manager (domain classification, intent recognition, search, small talk, sub-flows).
- Built NLP-side APIs over Astound's ontology and extended it to support inference-driven multi-turn clarification, establishing a separation of concerns between knowledge, lexicon, and grammar engineers.

Service

- Co-organizer, [Grounded User Simulation for Model Evaluation and Training](#), NeurIPS 2026 Workshop (Paris, Dec 2026).
- Organizer & judge, Sierra τ^2 -Bench Custom Track, [AgentX-AgentBeats](#) competition (Berkeley RDI, 2025–2026).

Education

University of Southern California

PhD, Computational Cognitive Neuroscience

Sep 2010 – Aug 2017
Los Angeles, CA

Dissertation: *Schema Architecture for Language–Vision Interactions: A Computational Cognitive Neuroscience Model of Language Use*. Advisor: Michael A. Arbib.

École Normale Supérieure

MS, Cognitive Science

Jun 2008 – Jun 2010
Paris, France

École Polytechnique

MS, Physics

Jun 2003 – Jun 2006
Palaiseau, France

Lycée Louis-le-Grand

Classes Préparatoires – Mathematics & Physics

Jun 2001 – Jun 2003
Paris, France

Selected Publications

τ^τ -Bench (Hyper- τ -Bench): An Environment for End-To-End, Realistic Agent Construction

Q. Shi, K. Dhandhan, K. Narasimhan, V. Barres.

arXiv:2609.04611, 2026

τ -Multilingual: Benchmarking Voice Agents Across Languages

S. Ray, E. dos Santos Paiva, R. Valenzuela, K. Narasimhan, K. Dhandhan, V. Barres.

arXiv:2609.35820, 2026

τ -Voice: Benchmarking Full-Duplex Voice Agents on Real-World Domains

V. Barres*, S. Ray*, K. Dhandhan*, K. Narasimhan.

arXiv:2603.13686, 2026. Accepted to ICML 2026.

τ -Knowledge: Evaluating Conversational Agents over Unstructured Knowledge

Q. Shi, A. Zytex, P. Razavi, K. Narasimhan, V. Barres.

arXiv:2603.04370, 2026. Accepted to ICML 2026.

τ^2 -Bench: Evaluating Conversational Agents in a Dual-Control Environment

V. Barres*, H. Dong*, S. Ray, X. Si, K. Narasimhan.

arXiv:2506.07982, 2025. ICML 2026 (Oral).

From Generating Answers to Building Explanations: Integrating Multi-Round RAG and Causal Modeling for Scientific QA

V. Barres, C. McFate, A. Kalyanpur, K. K. Saravanakumar, L. Moon, N. Seifu, A. Bautista-Castillo.
(Industry)

NAACL 2025

LLM-ARC: Enhancing LLMs with an Automated Reasoning Critic

A. Kalyanpur, K. Saravanakumar, V. Barres, J. Chu-Carroll, D. Melville, D. Ferrucci.

arXiv:2406.17663, 2024

Template Construction Grammar: From Visual Scene Description to Language Comprehension and Agrammatism

V. Barres, J. Lee.

Neuroinformatics 12(1), 2014

Synthetic Event-Related Potentials: A Computational Bridge between Neurolinguistic Models and Experiments

V. Barres, A. Simons, M. Arbib.

Neural Networks 37, 2013

* Equal contribution. Full list: victorbarres.github.io/publications/

Languages

English (fluent) · French (native) · Spanish (conversational)